Abhijeet Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.07
DAV Public School Dhori, Bokaro	Class XII (CBSE)	2024	93.60%
Carmel School Kargali, Bokaro	Class X (CBSE)	2022	94.80%

SKILLS

- **Programming Languages:** C++, C, Python, SQL

- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, PyTorch

- **Tools & Platforms:** Git, GitHub, LaTeX, Jupyter Notebook, Google Colab, VS Code

- **Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOPs), Database Management System, Machine Learning, Deep Learning

- **Soft Skills:** Team Collaboration, Leadership, Communication, Problem Solving, Time Management

PROJECTS

Image Decomposition & Bleedthrough Removal

(Mini-Project under Dr. Ashesh, Dept. of AI, IITH)

- Investigating bleedthrough removal in fluorescence microscopy with unavailable clean source channels
- Developing an iterative deep-learning framework for progressive removal of varying degradation severity levels
- Exploring self-supervised and decorrelation-based losses to suppress undesired cross-channel information leakage
- Integrating self-supervised denoising with image decomposition for robust real-world fluorescence microscopy data

Mini ML Framework and Autograd Engine

- Developed a scikit-learn-inspired ML framework with preprocessing, model selection, and dataset utilities.
- Implemented IRLS Logistic Regression, StandardScaler, Polynomial & Gaussian Basis Features, and K-Fold CV.
- Built an autograd engine with reverse-mode backpropagation, Linear, ReLU, Sigmoid, SGD, and Cross-Entropy.
- Trained OvR Logistic, Softmax Regression, and MLPs on Fashion-MNIST with hyperparameter tuning.

Deep Sequence Modeling for Sentiment Classification

- Implemented Vanilla RNNs and LSTMs from scratch with BPTT, gradient clipping, and trainable word embeddings.
- Built a Transformer Encoder with self-attention, positional encoding, residual connections, and LayerNorm.
- Developed Linear & Selective State Space Models (SSMs) for long-context sequence modeling up to 1024 tokens.
- Evaluated sequence models using SGD, Adam, and ablation studies on the IMDb sentiment classification dataset.

CNN-Based Image Classification

- Implemented a CNN from scratch with convolution, pooling, activation, flattening, and fully connected layers.
- Trained a PyTorch CNN on CIFAR-10 using SGD, Momentum, and RMSProp optimizers.
- Analyzed feature representations using activation maps, t-SNE, dropout, and batch normalization.

Neural Network Framework and Optimization Analysis

- Implemented Adaline and a modular MLP from scratch with custom forward and backpropagation algorithms.
- Developed configurable activation functions, optimizers, weight initialization, and L1/L2 regularization.
- Evaluated architectures, optimizers, and activation functions through extensive ablation studies.

Unsupervised Learning and Dimensionality Reduction

- Implemented Principal Component Analysis (PCA) from scratch for face reconstruction and dimensionality reduction.
- Built K-Means and Gaussian Mixture Models (GMM) from scratch and compared their clustering performance.

RELEVANT COURSES

- **AI :** Foundations of Machine Learning, Deep Learning, AI for Humanity, Introduction to Modern AI and Classical AI, Programming for Artificial Intelligence
- **CS :** Data Structures and Applications, Algorithms, DBMS, Discrete Mathematics, Introduction to Programming
- **Mathematics :** Probability and Random Variables, Optimization, Metric Spaces, Numerical Analysis, Calculus
- **EE :** Control Systems, Matrix Theory

POSITIONS OF RESPONSIBILITY

Manager, PR & Networking | Entrepreneurship Cell

- Ideated and led outreach for E-Merge 3.0, attracting 170+ participants from 15+ colleges across India.
- Organized flagship competitions Boardroom and Founders' Forum, boosting participant engagement.
- Built partnerships with startups, sponsors, and external organizations for E-Merge and E-Summit.

Assistant Manager, PR & Networking | Entrepreneurship Cell

- Pitched E-Merge to 20+ E-Cells and colleges, expanding outreach and driving cross-institutional participation.
- Supported planning, outreach, and execution of E-Cell initiatives through cross-functional collaboration.